

Document Status

Draft whitepaper for editorial refinement

Audience:

- defense product and platform teams
- imagery and geospatial analysts
- AI and data teams entering the Earth observation domain
- program leadership evaluating defense imagery intelligence capability

Scope:

- subject matter only
- not a procurement recommendation
- not a model benchmark paper
- not a legal or licensing opinion

Executive Summary

Remote sensing for defense imagery intelligence is not one technical discipline.

It is a stack of interdependent capabilities:

- sensor literacy
- imagery market literacy
- preprocessing and quality control
- temporal analysis
- object detection and recognition
- evidence and provenance management

The most important conclusion is simple:

high-performing models alone do not create a serious imagery intelligence capability.

What matters is the ability to answer, repeatedly and defensibly:

- what imagery exists
- what type of sensor collected it
- what level of interpretation it supports
- what changed between observations
- what the model inferred
- what a human analyst accepted
- how the full chain can be traced later

This is why the domain cannot be reduced to “get 30 cm imagery and run a detector.” The

actual operational problem is broader:

- optical imagery is intuitive but weather-limited
- SAR is persistent but interpretation-heavy
- commercial providers sell not just pixels but archive depth, tasking behavior, APIs, and licensing constraints
- change detection often creates more operational value than one-off fine-grained classification
- evidence lineage matters as much as raw model confidence

For a platform such as SHIELD, the correct architectural consequence is that detections,

chips, scene comparisons, and model outputs should enter the evidence plane first. They should not become canonical force-structure truth until reviewed and promoted through governed workflows.

Key Judgments

1. Spatial resolution is essential, but it is only one of four resolution concepts that matter in operational remote sensing: spatial, temporal, spectral, and radiometric.
2. Optical and SAR are complementary, not interchangeable. A serious monitoring stack should understand both.
3. Archive, tasking, repeat collection, delivery APIs, and licensing are core design constraints, not vendor-side details.
4. Change detection is one of the highest-value and hardest operational capabilities in imagery intelligence.
5. Object detection in overhead imagery is increasingly tractable on curated datasets, but fine-grained military recognition remains sensitive to resolution, geometry, and domain shift.
6. Public Earth observation systems teach cadence, harmonization, and time-series reasoning. Commercial systems provide the detail and responsiveness required for higher-value defense use cases.
7. The strongest production systems distinguish clearly between model hypotheses, analyst claim, and canonical intelligence truth.

1. Why This Domain Is Different

Remote sensing looks superficially similar to conventional computer vision: images come in, models analyze them, detections come out. That framing is misleading.

Defense imagery intelligence differs because:

- the sensor is remote and physically constrained
- collection geometry varies
- targets are small relative to scene size
- weather and cloud effects can dominate optical utility
- revisit cadence can matter more than image beauty
- archive history is often as important as fresh imagery
- evidence traceability matters downstream

The result is that imagery intelligence systems must be built as temporal and evidentiary systems, not just recognition systems.

2. The Core Vocabulary

2.1 Resolution Types

When teams say “resolution”, they usually mean spatial resolution alone. A publication on this topic should distinguish the four main resolution concepts explicitly.

Resolution type	Core question	Why it matters
Spatial detail visibility	How much ground does one pixel represent?	Determines geometric and small-object
Temporal ordering and	How often can the same place be observed?	Determines monitoring utility
Spectral discrimination	Which wavelength bands are captured?	Determines material environmental discrimination
Radiometric contrast, some	How finely are intensity differences recorded?	Affects subtle calibration, and analytical quality

Ground sample distance, or GSD, belongs to the spatial category. It is important, but it does not by itself tell an analyst what can be confidently identified.

2.2 Detection, Classification, Interpretation

Three questions must be kept separate:

- detection: is there an object here
- classification: what is it
- interpretation: what does it mean

This is more than terminology discipline. It prevents a common operational failure: confusing a model output with an intelligence conclusion.

3. Sensor Modalities

3.1 Optical Electro-Optical Imagery

Optical imagery remains the primary modality for human-readable scene interpretation and shape-based object recognition.

Strengths:

- intuitive for analysts
- strong fit for conventional computer vision
- useful for morphology-driven recognition
- strong for mapping, annotation, and evidentiary presentation

Weaknesses:

- cloud-limited
- daylight-dependent
- sensitive to haze, shadow, and off-nadir effects

For aircraft recognition, optical imagery is usually the primary source for family and subtype reasoning where sufficient detail exists.

3.2 Multispectral Imagery

Multispectral systems capture multiple wavelength bands beyond visible RGB, commonly including near-infrared and shortwave infrared.

Operational benefits:

- land-cover and water discrimination
- cloud and shadow handling
- environmental context
- stronger time-series normalization

Multispectral imagery is often more important to change detection, masking, and context than to fine-grained aircraft discrimination.

3.3 Hyperspectral Imagery

Hyperspectral systems capture many narrow bands and support material-oriented analysis.

This modality matters strategically, but it is not usually the first practical wedge for overhead aircraft monitoring. It becomes more relevant for:

- material or coating discrimination
- anomaly detection
- advanced environmental signatures

3.4 Synthetic Aperture Radar

SAR is one of the most important subjects for a serious monitoring capability.

Its practical strengths are decisive:

- all-weather collection
- day-night imaging
- useful revisit behavior
- strong fit for change detection
- coherence and interferometric applications

Its practical weaknesses are equally important:

- less intuitive interpretation for non-specialists
- speckle and geometric effects
- different modeling and preprocessing requirements

The operational lesson is not that SAR replaces optical imagery. The lesson is that weather-independent monitoring is a core requirement in real-world systems, and SAR is one of the main ways to satisfy it.

3.5 Sensor Comparison

Modality	Best for	Main limitation
Optical	Human-readable interpretation and shape recognition	Cloud and daylight dependence
Multispectral	Time series, masking, land-cover context	Lower detail for small objects
Hyperspectral	Material discrimination	More specialized and less commonly available
SAR	Persistent all-weather monitoring	Harder interpretation and different model stack
Thermal	Heat and night-related signals	Secondary for most aircraft workflows

4. Public Earth Observation Systems

Public systems are foundational because they teach cadence, harmonization, and temporal analysis at scale.

4.1 Sentinel-2

Copernicus describes Sentinel-2 as a high-resolution multispectral mission with 13 spectral bands: 4 at 10 m, 6 at 20 m, and 3 at 60 m, with a 290 km swath. This makes it one of the core public systems for land monitoring and time-series analysis. [1]

Best use:

- land and infrastructure context
- broad environmental change
- medium-resolution time series
- cloud-aware baseline building

Poor use:

- fine-grained military aircraft subtype recognition

4.2 Sentinel-1

Sentinel-1 is the major public SAR reference mission for many operational workflows. ESA and Copernicus position it as an all-weather, day-night SAR system with multiple ima

ging modes
and wide-area monitoring value. [2][3]

Best use:

- weather-independent monitoring
- persistent change analysis
- coherence-based workflows

Poor use:

- visually intuitive fine-grained aircraft interpretation without radar expertise

4.3 Landsat

USGS positions Landsat as a long-running moderate-resolution archive with continuous coverage reaching back to 1972, with current Landsat 8 and 9 systems delivering 30 m imagery. [4]

Best use:

- long historical baselines
- scientific time-series reasoning
- calibration-oriented thinking

Poor use:

- site-level aircraft recognition

4.4 Public Systems Summary

System	Modality	Typical role in workflow
Sentinel-2	Optical multispectral	Broad context, monitoring, time series
Sentinel-1	SAR	All-weather revisit and change
Landsat	Optical multispectral	Long historical baseline

5. Commercial Earth Observation Market

For publication purposes, it is important to explain that imagery providers differ not only by resolution, but by operating model.

5.1 What Providers Actually Sell

Most providers sell some combination of:

- archive access
- new tasking
- repeat monitoring
- mosaics or basemaps
- API-based delivery
- cloud-based search and ordering

That means customers are buying:

- access model
- latency
- coverage behavior

- licensing terms
- archive depth
- evidence usability

not just pixels.

5.2 Premium Optical Providers

Maxar / Vantor

As of May 12, 2026, Vantor public materials describe premium collection capacity including up to 15 revisits per day at the same location and approximately 3.5 million square kilometers per day at 30 cm resolution. [5]

Operationally, this is the premium-detail end of the market:

- high detail
- strong archive
- strong tasking relevance
- widely used in defense and mapping contexts

Airbus

Airbus public materials position Pléiades Neo as 30 cm imagery available through OneAtlas with archive and tasking access. Airbus also exposes mixed optical and radar access through the same broader platform posture. [6][7][8]

Operationally, Airbus matters as a major premium alternative with strong archive-plus-tasking behavior.

BlackSky

BlackSky's March 12, 2026 press release states that its Gen-3 system reached general availability and delivers 35 cm imagery. Its public product story emphasizes timely insight and integrated analytics. [9][10]

Operationally, BlackSky represents the monitoring-and-analytics end of the very-high-resolution optical market.

Planet High Resolution / SkySat

Planet documentation states that SkySat consists of 15 satellites, supports revisit up to 10 times daily for a location, and produces 4-band and panchromatic imagery sampled at 50 cm when orthorectified. [11]

Operationally, this is a strong reference for API-first, high-cadence, high-resolution tasking workflows.

5.3 Medium-Resolution Daily Monitoring

PlanetScope is important because it teaches a different market lesson.

Planet documentation states that PlanetScope achieved near-daily coverage over 1 and masses and exposes analysis-ready products for machine learning and time-series use. Planet commercial materials position mosaics and near-daily 3 m workflows as a key value layer. [12][13]

This matters because:

- medium resolution can be extremely valuable for temporal baselines
- cadence can dominate point detail for many monitoring tasks
- machine learning pipelines often depend on consistent time-series products

5.4 Commercial SAR Providers

ICEYE

ICEYE public product material emphasizes SAR as a change-detection-oriented modality and describes imaging modes with ground resolutions including 25 cm, 50 cm, and 1 m. [14]

Capella Space

Capella public materials position Spotlight imagery at 50 cm and expose both tasking and archive workflows. Their tasking documentation also reflects how modern SAR ordering and retasking are exposed through APIs and consoles. [15][16][17]

Umbra

Umbra public material highlights a 16 cm commercial SAR image release and emphasizes extreme detail in modern radar collection. [18]

5.5 Commercial Market Summary

Provider category	Typical strength	Strategic lesson
Premium optical detail matters	Fine detail and evidentiary interpretation	Use when shape d
High-cadence optical d looks matter	Monitoring and rapid revisit	Use when repeate
Medium-resolution daily and trend	Time series and broad monitoring	Use for baseline logic
Commercial SAR reliability	All-weather persistence	Use when optical is insufficient

6. Archive, Tasking, Monitoring, and Licensing

6.1 Archive

Archive imagery is previously collected imagery already accessible for search or purchase.

Why it matters:

- historical baselining
- prior-state comparison
- site history
- evaluation and model training

If a team ignores archive depth, it will often misunderstand the actual monitoring value of a provider.

6.2 Tasking

Tasking is the process of requesting future collection over an AOI and time window.

Typical tasking parameters include:

- AOI geometry
- acquisition window
- priority
- product type
- look-angle preferences
- cloud constraints

Tasking is how a passive imagery buyer becomes an active collection consumer.

6.3 Repeat Monitoring

Repeat monitoring turns collection into an operational feed-like behavior:

- recurring AOIs
- repeated checks
- event-driven collection
- new-imagery notifications

This is often more important to operational users than one-off exquisite tasking.

6.4 Basemaps and Mosaics

Basemaps are processed layers built from multiple scenes.

Strengths:

- broad context
- stable visualization
- regional interpretation

Limitation:

- they can obscure exact scene-level provenance

That makes them excellent context layers but weak substitutes for scene-specific evidence.

6.5 Licensing

Licensing affects:

- who may view imagery
- whether it can be shared with partners
- whether derived products can be redistributed
- whether tasked imagery eventually enters a broader archive

Planet's tasking documentation is a good example of why this matters: archive publication behavior and withhold options are part of the commercial model, not merely legal fine print.
[19][20]

7. Imagery Consumption Models

Imagery is usually consumed through more than one workflow at once.

7.1 Visual Interpretation

Analysts inspect scenes, compare dates, annotate findings, and build judgments.

7.2 GIS Integration

Scenes and footprints become layers in:

- desktop GIS
- web GIS
- geospatial intelligence workspaces

7.3 Time-Series Processing

Imagery is processed as a sequence rather than as isolated images.

7.4 Model Inference

Imagery is tiled, chipped, streamed, and passed through detection, segmentation, or change models.

7.5 Evidence Workflow

Outputs are stored together with:

- scene identifiers
- footprints
- timestamps
- chips
- model versions
- review state

This last mode is where imagery intelligence becomes a defensible platform capability.

8. Change Detection

8.1 Definition

Change detection is the identification of meaningful differences between observations at

different times.

The hard part is not simply finding difference. The hard part is distinguishing:

- real change on the ground
- apparent change caused by collection differences

8.2 Main Types

Type	Best for	Main weakness
Pixel-level	Surface or regional difference	Low semantic precision
Object-level detection	Appearance, disappearance, count change	Depends on reliable data
Semantic requirements	Typed change with meaning	Harder data and model
Time-series normalization demands	Trend and pattern analysis	Stronger storage and

8.3 Why It Is Hard

False change commonly comes from:

- registration error
- illumination changes
- cloud and shadow
- atmospheric variation
- off-nadir geometry differences
- modality mismatch
- SAR-specific speckle and geometry behavior

8.4 Preprocessing Requirements

Serious change detection usually requires:

- geometric co-registration
- radiometric normalization
- cloud and shadow masking
- valid-pixel handling
- cross-sensor harmonization where necessary

This is why analysis-ready products are strategically important: they reduce temporal instability in the input stack.

8.5 Current State of the Field

Recent review literature shows a field that is advancing quickly but is still not commoditized in difficult operational settings. Major method families include classical differencing, feature-based methods, Siamese CNNs, transformer-based architectures, and hybrid semantic change models. [21]

The key takeaway is that operational multi-source military change detection remains hard, especially when data quality varies.

9. Object Detection in Overhead Imagery

9.1 Why It Is Different from Natural-Image Detection

Overhead imagery differs because:

- scenes are large and sparse
- objects are small
- orientation varies
- class frequencies are imbalanced
- visual context differs from natural-image benchmarks

9.2 Main Analysis Tasks

Task	Question answered	Typical output
-----	-----	-----
Detection	Is an object present?	box, mask, score
Classification	What broad thing is it?	role or family label
Fine-grained recognition	Which subtype is it?	subtype hypothesis
Change detection	What changed between dates?	delta object or regio
n Interpretation	What does it mean?	analyst claim

9.3 Common Model Families

Representative families include:

- one-stage detectors such as YOLO variants
- two-stage detectors such as Faster R-CNN
- transformer-based detectors
- oriented object detectors

The choice of family still matters, but operational success depends heavily on data and deployment conditions.

9.4 What Current Research Suggests

A 2024 comparative review on aircraft detection in satellite imagery reported strong performance from YOLO-family models in the evaluated setting, with YOLOv5 leading the tested methods in that study. [22]

This is useful, but it should not be over-generalized across:

- all providers
- all GSD bands
- all military classes
- all deployment conditions

9.5 The Real Difficulty

Real-world failure modes include:

- provider-driven domain shift
- rare-class scarcity
- confusing family-level morphology
- occlusion and shelter effects
- off-angle views
- poor calibration of confidence

This is why good benchmark mAP does not automatically become good operational trust.

10. Fine-Grained Aircraft Recognition

Fine-grained recognition is the next difficulty layer beyond detection.

It is hard because:

- many variants differ by subtle shape cues
- the overhead perspective removes some distinguishing features
- shadows and orientation distort the evidence
- subtypes can be visually close even at high quality

The correct operational posture is hierarchical:

- object present
- broad role
- family
- subtype only when supported

This protects the system from false precision.

11. Data and Benchmarks

Public datasets are useful for orientation, benchmarking, and initial experimentation, but they are not a substitute for deployment-representative evaluation.

11.1 xView

xView is one of the major overhead object detection references, with more than one million annotated objects across 60 classes. [23]

11.2 RarePlanes

RarePlanes is an aircraft-focused benchmark designed in part to explore real-plus-synthetic learning strategies for overhead imagery. [24]

11.3 FAIR1M

FAIR1M is a large fine-grained remote sensing benchmark with more than one million instances and strong relevance to oriented detection and category richness. [25]

11.4 Benchmark Lesson

Benchmarks are necessary, but operational evaluation must also account for:

- provider mix
- regional mix
- licensing realities
- cloud and geometry variation
- long-tail class behavior

12. Foundation Models

Foundation models are now part of the remote sensing landscape, but they do not eliminate the need for domain-specific engineering.

NASA technical reporting on Prithvi-E0-2.0 describes a geospatial foundation model trained on 4.2 million global time-series samples from harmonized Landsat and Sentinel-2 data. NASA public reporting in 2026 also highlighted on-orbit deployment as a milestone for geospatial foundation models. [26][27][28]

These models matter because they may improve:

- transfer learning
- label efficiency
- multi-temporal reasoning
- general representation quality

But they do not remove the need for:

- target ontology
- deployment-specific evaluation
- provenance handling
- confidence calibration

13. Implications for SHIELD and Similar Platforms

For a platform such as SHIELD, the most important architectural principle is: model outputs should enter as evidence before they become truth.

That implies a pipeline like:

1. scene registered
2. derived chips and detections created
3. compare products generated
4. analyst annotations attached
5. claims reviewed
6. only approved conclusions promoted into canonical intelligence structures

13.1 Recommended Evidence Objects

A platform should distinguish at least:

- source asset
- scene
- footprint
- derived chip
- detection
- compare result
- annotation
- claim
- promotion decision

13.2 Why This Separation Matters

Without this separation, teams tend to:

- over-trust model outputs
- lose scene lineage
- blur analyst judgment and machine inference
- weaken auditability

With this separation, the platform becomes governable.

14. Common Failure Modes

The most common strategic and technical mistakes in this domain are:

1. treating nominal GSD as certainty
2. ignoring archive depth
3. thinking only in optical terms
4. overfitting to benchmarks
5. failing to normalize time series properly
6. storing detections without evidence lineage
7. confusing basemap context with scene-level evidence

15. Conclusion

Remote sensing for defense imagery intelligence is a systems problem.

The durable competitive asset is not only a model checkpoint. It is the operational stack that understands:

- sensor behavior
- commercial imagery supply
- temporal baselines
- object recognition limits
- change-detection workflows
- evidentiary promotion rules

Any serious whitepaper or product strategy in this space should make that explicit.

Glossary

AOI

area of interest

Archive imagery

imagery already collected and searchable or orderable from a provider catalog

Basemap

a harmonized visualization layer built from multiple scenes

Change detection

identification of meaningful differences across time

Chip

a cropped image region, often centered on a target or candidate target

GSD

ground sample distance, the ground size represented by one pixel

Multispectral

imagery with multiple broad wavelength bands

Off-nadir

collection at an angle away from straight-down viewing geometry

SAR

synthetic aperture radar, an active sensing modality capable of day-night, all-weather imaging

Scene
a single collected image product with acquisition identity and metadata

Tasking
requesting a future collection from a provider

Time series
a sequence of observations over time for the same place or AOI

References

[1] Copernicus, “Sentinel-2: high-resolution and multispectral.”
<https://www.copernicus.eu/en/media/images/sentinel-2-high-resolution-and-multispectral>

[2] Copernicus Data Space, “Sentinel-1.”
<https://dataspace.copernicus.eu/data-collections/sentinel-data/sentinel-1>

[3] ESA, “Sentinel-1.” <https://www.esa.int/Sentinel-1>

[4] USGS, “The Landsat Program.” <https://eros.usgs.gov/earthshots/the-landsat-program>

[5] Vantor, “Constellation.” <https://vantor.com/company/constellation/>

[6] Airbus, “Satellite imagery.”
<https://www.airbus.com/en/products-services/space/earth-observation/satellite-imagery>

[7] Airbus, “Pléiades Neo: Native 30cm High-Resolution Satellite Imagery.”
<https://space-solutions.airbus.com/imagery/pleiades-neo/>

[8] Airbus, “How to order Pléiades Neo.”
<https://space-solutions.airbus.com/imagery/how-to-order-imagery-and-data/how-to-order-pleiades-neo/>

[9] BlackSky, “BlackSky Successfully Commissions Gen-3 in Less Than One Week Opening Gen-3 General Availability for Global Customer Base,” March 12, 2026.
<https://ir.blacksky.com/news-events/press-releases/news-details/2026/BlackSky-Successfully-Commissions-Gen-3-in-Less-Than-One-Week-Opening-Gen-3-General-Availability-for-Global-Customer-Base/default.aspx>

[10] BlackSky, “Gen-3.” <https://www.blacksky.com/gen-3/>

[11] Planet Documentation, “SkySat.” <https://docs.planet.com/data/imagery/skysat/>

[12] Planet Documentation, “PlanetScope.” <https://docs.planet.com/data/imagery/planet-scope/>

[13] Planet, “Planet Mosaics.” <https://www.planet.com/products/mosaics/>

[14] ICEYE, “Imaging modes.” <https://www.iceye.com/sar-imagery>

[15] Capella Space, “SAR Data.” <https://www.capellaspace.com/solution/sar-data>

[16] Capella Space Docs, “Tasking Requests.”
<https://docs.capellaspace.com/constellation-tasking/tasking-requests/>

[17] Capella Support, “Creating a single task.”

<https://support.capellaspace.com/creating-a-single-task>

[18] Umbra, “Umbra Generates the Highest Resolution Commercial Satellite Image Ever Released.”
<https://umbra.space/blog/umbra-generates-the-highest-resolution-commercial-satellite-image-ever-released/>

[19] Planet Documentation, “Tasking.” <https://docs.planet.com/develop/apis/tasking/>

[20] Planet Documentation, “Archive tasking.”
https://docs.planet.com/develop/apis/tasking/archive_tasking/

[21] C. B. de Oliveira et al., “Remote Sensing Image Change Detection: A Review,” *Remote Sensing* 15(8):2092, 2023. <https://www.mdpi.com/2072-4292/15/8/2092>

[22] A. Abusham et al., “Deep Learning Models for Aircraft Detection in Satellite Imagery: A Comparative Analysis,” *Remote Sensing* 16(24):4715, 2024.
<https://www.mdpi.com/2072-4292/16/24/4715>

[23] xView dataset. <https://xviewdataset.org/>

[24] RarePlanes dataset. <https://registry.opendata.aws/rareplanes/>

[25] FAIR1M dataset paper. <https://arxiv.org/abs/2103.05569>

[26] NASA NTRS, “Prithvi-E0-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation Applications.” <https://ntrs.nasa.gov/citations/20240015391>

[27] NASA Science, “AI foundation model in orbit.”
<https://science.nasa.gov/science-research/ai-foundation-model-in-orbit/>

[28] NASA SC24, “Foundation Models for Generalist Geospatial Artificial Intelligence.”
<https://www.nas.nasa.gov/SC24/research/project27.php>